

Linux Commands

A working reference for the commands you actually type at a shell. Distro-agnostic where it matters, with notes on the `apt` vs `dnf` split for package management.

Navigation & Files

Command	Action
<code>pwd</code>	print the current working directory
<code>ls -lah</code>	long listing, a ll files, h uman-readable sizes
<code>cd -</code>	jump back to the previous directory
<code>cp -r src dst</code>	copy recursively (directories)
<code>mv old new</code>	move or rename
<code>rm -r dir</code>	remove a directory and its contents
<code>mkdir -p a/b/c</code>	create nested directories in one shot
<code>touch file</code>	create an empty file / update its timestamp
<code>ln -s target link</code>	create a symbolic link
<code>tree -L 2</code>	show the directory tree, 2 levels deep

rm -rf is unforgiving. There is no trash can. Double-check the path before running `rm -rf`, and never run it with a variable that could be empty (`rm -rf "$DIR/"` deletes `/` if `$DIR` is unset).

Viewing & Searching

Command	Action
<code>cat file</code>	dump a whole file to stdout
<code>less file</code>	page through a file (<code>q</code> to quit, <code>/</code> to search)
<code>head -n 20 file</code>	first 20 lines
<code>tail -f log</code>	follow a file as it grows — great for logs
<code>grep -rin 'text' .</code>	recursive, case-insensitive, with line numbers
<code>find . -name '*.log'</code>	find files by name pattern
<code>find . -mtime -1</code>	files modified in the last day
<code>wc -l file</code>	count lines
<code>`sort file`</code>	<code>uniq -c`</code>

Permissions & Ownership

Command	Action
<code>chmod +x script.sh</code>	make a file executable
<code>chmod 644 file</code>	owner read/write, group + others read
<code>chmod 755 dir</code>	owner all, group + others read/execute
<code>chown user:group file</code>	change owner and group
<code>chmod -R 750 dir</code>	apply recursively
<code>sudo !!</code>	re-run the previous command with sudo

Octal in three digits: read = 4, write = 2, execute = 1. Add them per role (owner, group, other). `7` = `rw``x`, `6` = `rw`-, `5` = `r`-`x`.

Processes & System

Command	Action
<code>`ps aux`</code>	<code>grep node`</code>
<code>top</code>	live process/CPU view (<code>htop</code> if installed)
<code>kill -9 PID</code>	force-kill a process by PID
<code>pkill -f pattern</code>	kill by matching the full command line
<code>df -h</code>	disk space by filesystem, human-readable
<code>du -sh *</code>	size of each item in the current directory
<code>free -h</code>	memory usage
<code>systemctl status nginx</code>	check a service's state (systemd)
<code>journalctl -u nginx -f</code>	follow a service's logs

Networking

Command	Action
<code>curl -I https://site</code>	fetch just the response headers
<code>`curl -fsSL url`</code>	<code>sh`</code>
<code>wget url</code>	download a file
<code>ssh user@host</code>	open a remote shell
<code>scp file user@host:/path</code>	copy a file over SSH
<code>ss -tulpn</code>	listening ports and their processes
<code>ping -c 4 host</code>	send 4 ICMP probes
<code>dig +short example.com</code>	resolve a domain, terse output
<code>ip a</code>	show network interfaces and addresses

Archives & Packages

Command	Action
<code>tar czf out.tgz dir/</code>	create a gzipped tarball
<code>tar xzf out.tgz</code>	extract a gzipped tarball
<code>tar tzf out.tgz</code>	list contents without extracting
<code>zip -r out.zip dir/</code>	create a zip archive
<code>unzip out.zip</code>	extract a zip archive

Package managers — Debian/Ubuntu vs Fedora/RHEL

```
# Debian / Ubuntu (apt)
sudo apt update && sudo apt upgrade
sudo apt install <pkg>
sudo apt remove <pkg>

# Fedora / RHEL / Rocky (dnf)
sudo dnf check-update
sudo dnf install <pkg>
sudo dnf remove <pkg>
```